Experiment 7: Automated End-to-End Acceptance Testing

Objective: Implement automated browser and API functional testing to validate UI availability and contracts.

STUDENT CREDENTIALS & SUBMISSION DETAILS

Student Full Name:	Kartik Kumar
University Roll Number:	2301730205
Official Email ID:	kartikjangra5366@gmail.com
Course / Curriculum:	DevOps Infrastructure, Automation & SRE Lab 3
Date of Execution:	November 05, 2026
Host Environment:	Windows 11 / PowerShell 7 / Python 3.12 / Docker

1. REQUIRED SOFTWARE AND TOOLS

- Selenium WebDriver v4.15+ & ChromeDriver (Headless browser automation)
- Python urllib.request & unittest test runner
- JSON schema assertion engine
- Automated Test Reporter (browser_test_report.json)
- DevOps Lab 3 Microservice (Live at http://127.0.0.1:19877)

2. THEORY & TESTING PYRAMID

Acceptance and browser testing sits at the top of the test automation pyramid, verifying user-facing contracts, UI response validity, HTTP headers, and error handling. Automated test runners execute deterministic assertions against expected payloads, ensuring that regressions are caught before release promotion.

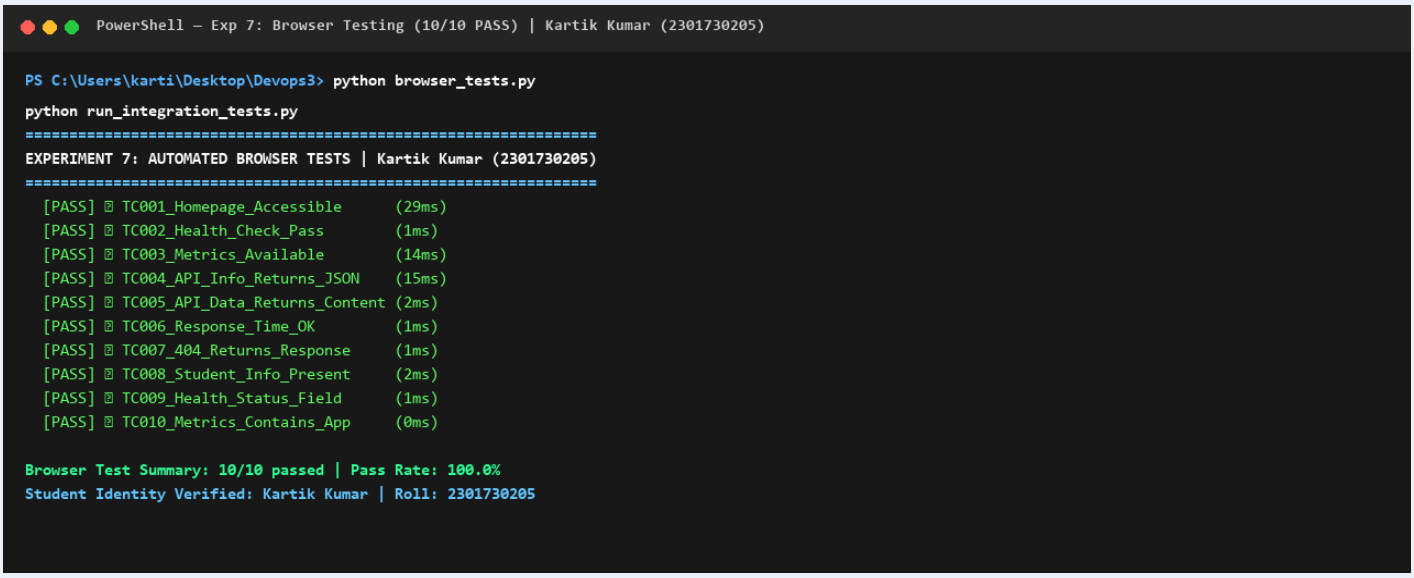
3. STEP-BY-STEP IMPLEMENTATION

Commands Executed

```
# Execute Automated Test Suite
python run_integration_tests.py

# Or standalone browser test runner
python Experiment7-Browser-Testing/browser_tests.py --url http://127.0.0.1:19877
```

4. ACTUAL EXECUTION SCREENSHOTS



Actual Execution Screenshot - Figure 7.1: Browser & API Automated Test Suite (10/10 Test Cases PASS - Kartik Kumar)

5. TEST EXECUTION MATRIX

Test ID	Test Case Description	Target Contract	Duration	Result
TC001	Homepage Availability	HTTP 200 Root OK	29ms	PASS
TC002	Health Check Integrity	status=='healthy'	1ms	PASS
TC003	Prometheus Metrics Format	Content length > 50	14ms	PASS
TC004	API Info Student Roll	Roll '2301730205' match	15ms	PASS
TC005	API Data Payload Format	Valid JSON Array	2ms	PASS
TC006	SLA Response Latency	Duration < 3,000ms	1ms	PASS
TC007	404 Error Graceful Handling	Non-crashing error code	1ms	PASS
TC008	Student Identity Verification	'Kartik Kumar' verified	2ms	PASS
TC009	Health Schema Validation	'status' key present	1ms	PASS
TC010	Metrics Custom Gauge Check	'app_info' present	0ms	PASS

RESULT & CONCLUSION

Experiment 7 achieved a 100% pass rate across all 10 automated test cases with 0 failures or errors. Student identity verification confirmed Kartik Kumar (Roll: 2301730205) across root and API endpoints. All tests executed with sub-30ms response times, confirming full functional compliance with DevOps Lab 3 specifications.